

# Ecuaciones de evolución / Equilibrio

Principales:

· Ecuaciones de movimiento

$$u(t): \frac{d^2u}{dt^2} = \frac{F(u)}{m} \quad F=0$$

· Ecuaciones de Conservación

$$\frac{dU}{dt} = m \cdot c_p \cdot \frac{dT}{dt} = Q(T) \quad Q=0$$

$$\frac{dU}{dt} = F(u) + f$$

Fuerzas      Calores

$f=0$ ; Autónomo

$f \neq 0$ ; No autónomo

$$\frac{d^2u}{dt^2} = F(u) \quad \frac{du}{dt} = v \quad \frac{dv}{dt} = \frac{d^2u}{dt^2} \quad \frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ F(u) \end{pmatrix} = \begin{pmatrix} F_1(u,v) \\ F_2(u,v) \end{pmatrix} \rightarrow \begin{pmatrix} v \\ F(u) \end{pmatrix}$$

$$\vec{u} = \begin{pmatrix} u \\ v \end{pmatrix} \quad \frac{d\vec{u}}{dt} = \begin{bmatrix} 0 & 1 \\ -k & 0 \end{bmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} 0 \\ f \end{pmatrix} \rightarrow \frac{d\vec{u}}{dt} = [K] \vec{u} + \vec{f} \quad \text{Sistema lineal}$$

Ejemplo Termico

$h, T_{\infty}$

$$U = m \cdot c_p \cdot T(t) \quad \frac{dU}{dt} = \sum Q \quad m c_p \frac{dT}{dt} = h \cdot A \cdot (T_{\infty} - T) \quad \frac{dT}{dt} = \left( -\frac{hA}{m c_p} \right) T + \frac{hA}{m c_p} T_{\infty}$$

$$\frac{dT}{dt} = -\alpha T + q \quad \frac{dT}{dt} = F(T) + q$$

$T_{\infty 1}, h_1$        $T_{\infty 2}, h_2$

$T_1$   
 $m_1 c_p$

$T_2$   
 $m_2 c_p$

$\longleftrightarrow$   
 $C_k$

$m_1 T = m_1 c_p$

$$m_1 T \cdot \frac{dT_1}{dt} = h_1 A_1 (T_{\infty 1} - T_1) + C_k (T_2 - T_1)$$

Repasar

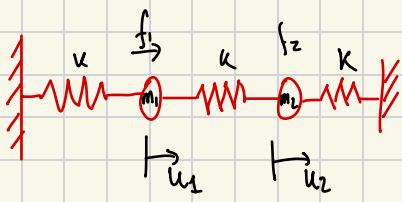
$$m_2 T \cdot \frac{dT_2}{dt} = h_2 A_2 (T_{\infty 2} - T_2) + C_k (T_1 - T_2)$$

$$\rightarrow \begin{bmatrix} m_1 T & 0 \\ 0 & m_2 T \end{bmatrix} \begin{pmatrix} \frac{dT_1}{dt} \\ \frac{dT_2}{dt} \end{pmatrix} = \begin{bmatrix} -h_1 A_1 - C_k & C_k \\ C_k & -h_2 A_2 - C_k \end{bmatrix} \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} + \begin{pmatrix} h_1 A_1 T_{\infty 1} \\ h_2 A_2 T_{\infty 2} \end{pmatrix}$$

IM       $\frac{d}{dt} \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = [K] \vec{T} + \vec{Q}$

$IM \frac{d}{dt} \vec{T} = [K] \vec{T} + \vec{Q}$

difusivo  
 ↓  
 Termico



$$F = -kx$$

$$F = ma$$

$$m_1 \frac{d^2 u_1}{dt^2} = -k u_1 + k(u_2 - u_1) + f_1$$

$$m_2 \frac{d^2 u_2}{dt^2} = -k u_2 + k(u_1 - u_2) + f_2$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{pmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{pmatrix} = \begin{bmatrix} -2k & k \\ k & -2k \end{bmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

$$M \ddot{\vec{U}} = K \vec{U} + \vec{F}$$

Oscilatorio

[Mechanica]

$$q'' + q' = 0$$

$$\frac{dT}{dt} = -\alpha T \quad T = T(0)e^{-\alpha t}$$

I

II

$$m \frac{d^2 u}{dt^2} = -ku \quad \ddot{u} + \frac{k}{m} u = 0$$

$$\frac{dT}{T} = -\alpha dt \quad \ln(T) = -\alpha t \quad T = e^{-\alpha t}$$

$$T = e^{-\alpha t}$$

$$\lambda^2 + \frac{k}{m} = 0 \quad \lambda(\lambda + \frac{k}{m}) \neq 0$$

$$\lambda = \pm \sqrt{\frac{k}{m}} \quad u(t) = e^{\lambda t} \quad u(t) = e^{-\lambda t}$$

$$u(t) = C_1 e^{\lambda t} + C_2 e^{-\lambda t} \quad \lambda = \pm i \sqrt{\frac{k}{m}}$$

$$u(t) = C_1 e^{i\lambda t} + C_2 e^{-i\lambda t}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$m \frac{d^2 u}{dt^2}$$